Phase 1: Critical Fixes Before Using in Production

Act as a senior Python workflow engineer and production-readiness auditor.

I need you to implement the following Phase 1 critical fixes in the project. These issues must be fixed before the software is considered usable in production because they can break the workflow, cause duplicate work, or create pipeline dead ends.

Critical fixes to implement:

#	Issue	Required Fix
1	Empty logging module	Implement a proper logging system with both file and console handlers
2	No duplicate upload prevention	Implement `upload_db.py` and add a pre-upload check before any upload happens
3	In-memory-only render job state	Persist render job state to disk so render progress/ status survives server restarts
4	`run_all_placeholder` dead end	Wire it to the full workflow or delegate it properly to `workflow_all.py`
5	Duplicate `render_softness_sigma` function	Delete the duplicate function and keep only one correct implementation

For each fix:

1. Locate the relevant files and modules.
2. Explain the current issue briefly.
3. Apply the fix directly in the code.
4. Make sure the fix does not break existing functionality.
5. Add clear error handling where needed.
6. Add or update tests if possible.
7. Verify that the pipeline can continue, retry, or recover instead of reaching a dead end.
8. Summarize the files changed and what was changed.

Important workflow requirement:

The pipeline must not force me to restart everything from the beginning when one step fails. The system should support safe retry, resume, skip, or recovery wherever possible.

Expected output:

1. Summary of all Phase 1 fixes completed
2. Files modified
3. Explanation of each fix

4. Any new files created
5. Any tests added or updated
6. Any remaining risks or follow-up tasks
7. How to run and verify the fixes

Treat this as a production-readiness patch, not just a quick bug fix.